

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I M.DEVI SRILATHA(192525236) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

1. Type-1 vs Type-2 Hypervisors

Type-1 Hypervisor:

A Type-1 hypervisor runs directly on the physical hardware without requiring a normal host operating system. Examples are VMware ESXi, Microsoft Hyper-V, and Xen.

Type-2 Hypervisor:

A Type-2 hypervisor runs on top of a host operating system. Examples include VMware Workstation and Oracle VirtualBox.

Factor	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High	Comparatively lower
Security	More secure	Less secure due to host OS
Management	Suitable for large datacenters	Easy for small environments
Hardware access	Direct	Through host OS
Use	Cloud and enterprise servers	Testing and personal use

Analysis:

For mission-critical cloud applications, performance and security are very important. Type-1 hypervisors have less overhead because they directly communicate with hardware. They also provide better isolation between virtual machines.

Manageability Considerations:

- Type-1 hypervisors integrate with centralized management consoles such as vCenter, enabling patching, monitoring, and live migration across the entire cluster from a single interface.
- Licensing and support costs are typically higher for Type-1 hypervisors, but this is offset by reduced downtime and stronger operational control in production environments.
- Type-2 hypervisors remain useful for development, training labs, and proof-of-concept work where isolation from production hardware is not required.

Conclusion:

I would choose a **Type-1 hypervisor** for a cloud datacenter because it provides **better performance, stronger security, reliability, and centralized management**.

2. Workload Isolation in Cloud Virtualization

A cloud provider may have finance, healthcare, and student applications running on the same physical server. Virtualization creates separate **Virtual Machines (VMs)** for each workload.

For example:

Physical Server → Hypervisor → Finance VM | Healthcare VM | Student Portal VM

The hypervisor allocates separate CPU, memory, storage, and network resources to each VM. Therefore, one workload cannot directly access another workload's resources.

How virtualization provides isolation:

- Each VM has its own virtual CPU and memory.
- Virtual disks are separated between VMs.
- Virtual networks can be configured separately.
- Hypervisor controls access to hardware resources.
- Security policies can be applied independently.

Two major attack surfaces:

1. Hypervisor attack:

If an attacker exploits a vulnerability in the hypervisor, it may affect multiple VMs. This is called a **VM escape** type of risk.

2. Virtual network attack:

Attackers may try to exploit incorrectly configured virtual switches, network connections, or security rules to communicate with another workload.

Additional Mitigation Measures:

- Apply micro-segmentation so that even workloads on the same host cannot communicate unless explicitly permitted.
- Keep the hypervisor patched and regularly audited, since most VM escape vulnerabilities are fixed quickly once disclosed.
- Use VM introspection and behavioural monitoring tools to detect unusual activity inside guest VMs.
- Apply workload-specific firewall and security-group rules so finance, healthcare, and student traffic remain logically separated even on shared infrastructure.

Conclusion:

Virtualization provides strong workload isolation, but the hypervisor and virtual networking must be properly secured.

3. VM Host Utilization Analysis

Given values:

- **CPU = 92%**
- **Memory = 88%**
- **Storage I/O latency = 35 ms**
- **Average VM boot time = 170 seconds**

These values indicate that the host is **highly utilized**.

Likely bottlenecks:

1. **CPU Bottleneck:**
CPU utilization of 92% is very high. VMs may have to wait for CPU resources, resulting in slower application performance.
2. **Memory Bottleneck:**
Memory utilization of 88% indicates that available RAM is nearly exhausted. This can cause memory contention and swapping.
3. **Storage I/O Bottleneck:**
35 ms storage latency is relatively high for many VM workloads. Slow disk operations can increase VM boot time and application response time.
4. **High VM Boot Time:**
An average boot time of 170 seconds suggests that CPU, memory, and especially storage I/O are affecting VM startup performance.

Corrective actions:

- Add more CPU resources or distribute VMs to other hosts.
- Increase physical RAM.
- Migrate VMs using **live migration** to less-loaded hosts.
- Upgrade storage to faster SSD/NVMe-based storage.
- Use storage caching where appropriate.
- Reduce unnecessary VM resource allocation.
- Monitor the host continuously.

Preventive Monitoring Strategy:

- Set proactive utilization thresholds (for example, alert at 75% CPU/memory) so action can be taken before performance degrades.
- Use historical trend data for capacity planning to forecast when additional hardware will be required.
- Apply predictive analytics or AI-based monitoring to anticipate spikes before they cause bottlenecks.

- Schedule regular maintenance windows to rebalance VM placement across the host cluster.

Conclusion:

The main problem is **overall resource overutilization**, with CPU and memory pressure and high storage latency contributing to the long VM boot time.

4. Software Defined Datacenter vs Traditional Datacenter

A **Software Defined Datacenter (SDDC)** uses software and virtualization to manage computing, storage, and networking resources. In a traditional datacenter, these resources are mainly configured and managed using physical hardware.

Feature	Traditional Datacenter	SDDC
Compute	Physical servers	Virtual machines
Storage	Physical storage devices	Software-defined storage
Network	Physical switches/routers	Software-defined networking
Management	Mostly manual	Software-based/automated
Scalability	Slower	Faster
Agility	Low	High

Compute Virtualization:

Servers can be divided into multiple VMs. New VMs can be created quickly based on application requirements.

Storage Virtualization:

Different physical storage resources can be combined and managed as a common storage pool. Storage can be allocated dynamically.

Network Virtualization:

Virtual networks can be created without changing physical network infrastructure. Network policies can also be changed using software.

How SDDC improves agility:

- Faster resource provisioning.
- Automated infrastructure management.
- Easy scaling up and down.
- Better resource utilization.
- Faster deployment of applications.
- Reduced dependency on manual hardware configuration.

Real-World Example:

- A traditional datacenter may take several weeks to provision a new server, including hardware procurement, racking, and manual network configuration.
- In an SDDC, the same request can be fulfilled in minutes through a self-service portal, since compute, storage, and network are all defined and provisioned in software.

- Organizations adopting SDDC platforms such as VMware Cloud Foundation report significantly faster application rollout and lower operational overhead as a direct result of automation.

Conclusion:

SDDC provides greater flexibility, automation, scalability, and faster resource provisioning compared with traditional datacent

5. Multi-Cloud Federation and Identity Mismatch

In a multi-cloud environment, an organization may use services from different cloud providers. Each provider may use different **identity formats, authentication systems, roles, and access policies**.

For example, a user may have one identity in Cloud Provider A but a different identity in Cloud Provider B. This creates an **identity mismatch** and can cause authentication and authorization failures.

Problems caused by identity mismatch:

- Users may need multiple accounts.
- Authentication may fail between providers.
- Roles and permissions may not match.
- User information may be duplicated.
- Privacy risks can increase.

Secure identity design:

A federation-based approach can be used.

User → Identity Provider (IdP) → Cloud Provider A / Cloud Provider B

The organization can use a central **Identity Provider (IdP)** to authenticate users. The identity provider can securely provide authentication information to different cloud providers using federation standards such as **SAML or OpenID Connect**.

Security measures:

1. Use a centralized Identity Provider.
2. Use Single Sign-On (SSO).
3. Apply Multi-Factor Authentication (MFA).
4. Use role-based access control (RBAC).
5. Follow the principle of least privilege.
6. Encrypt identity information during transmission.
7. Share only the minimum required user information.
8. Use short-lived authentication tokens.
9. Maintain logs and monitor access.

Privacy design:

Only necessary user attributes should be shared with cloud providers. Sensitive information should not be unnecessarily transferred between providers.

Additional Federation Standards and Practices:

10. OAuth 2.0 can be used alongside SAML/OpenID Connect to secure API-level access between federated cloud services.
11. SCIM (System for Cross-domain Identity Management) can automate user provisioning and de-provisioning consistently across multiple cloud providers.
12. Maintaining a unified, cross-cloud audit trail helps detect anomalous authentication attempts that might indicate identity mismatch exploitation.
13. Periodic access reviews ensure that federated identities do not accumulate excessive or outdated permissions over time.

Conclusion:

Cloud federation with a trusted Identity Provider, SSO, MFA, secure tokens, and minimum data sharing can solve identity mismatches while providing **secure authentication, authorization, and privacy across multiple clouds**.

